DEMO / SYNTHETIC: Powder bed preheating and laser power effects on porosity and residual stress in L-PBF Ti6Al4V

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Abstract

DEMO / SYNTHETIC TEXT generated to test the AM-KG pipeline; the numbers are not real research results. Ti6Al4V parts were fabricated by L-PBF. A preheating temperature of 500 °C reduced the residual stress, and the laser power influenced porosity.

2. Experimental setup

Ti6Al4V cubes were produced by L-PBF with a laser power of 280 W, a scanning speed of 1.2 m/s, a layer thickness of 60 µm and a preheating temperature of 500 °C. Porosity was quantified with micro-CT and residual stress was measured by XRD.

3. Results and discussion

Lower laser power increased lack of fusion porosity. The preheating temperature decreased the residual stress from 450 MPa to 120 MPa. The relative density of the cubes was 99.8 %. Preheating promoted the decomposition of α' martensite.

4. Conclusions

Preheating temperature affected the residual stress of L-PBF Ti6Al4V.